Development Bank of India was established in 1990 as a wholly owned subsidiary of the Industrial Development Bank of India with the aim to aid in the growth and development of micro, small and medium-scale enterprises (MSME). Currently, the ownership is held by 34 Government of India owned/controlled institutions.

66. **Assertion (A) :** India does not export natural rubber.

Reason (R) : About 97% of India's demand for natural rubber is met from domestic production.

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true but R is NOT the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

Ans. (a): India does not export natural rubber as most of the natural rubber produced in India is been consumed within the country. Therefore most of the demand of natural rubber within the country is fulfilled by domestic production.

67. **Assertion (A) :** For the first time, India had no trade deficit in the year 2002-03

Reason (R) : For the first time, India's exports crossed worth $ 50 billion in the year 2002-03.

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true but R is NOT the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

Ans. (d) : Assertion is not correct as the trade deficit fell from 2000-01 to 2002-03 but there was trade deficit in 2002-2003.

68. **Consider the following statements:**

1. **Regarding the procurement of food grains, Government of India follows a procurement target rather than an open-ended procurement policy.**
2. **Government of India announces minimum support prices only for cereals.**
3. **For distribution under Targeted Public Distribution System (TPDS), wheat and rice are issued by the Government of India at uniform central issue prices to the States/Union Territories.**

Which of the statements given above is/are correct?

(a) 1 and 2 (b) 2 only

(c) 1 and 3 (d) 3 only

Ans. (c) : The Government's price policy for agricultural commodities seeks to ensure remunerative prices to the growers for their produce with a view to encourage higher investment and production and to safeguard the interest of consumers by making available supplies at reasonable prices with low cost of intermediation. Government announces, Minimum Support Prices (MSP) for 25 major agricultural commodities each year in both the Crop seasons after taking into account the recommendations of the Commission for Agricultural Costs and Prices (CACP). Hence statement 2 is not correct. Rest of the statements is correct.

69. **Consider the following statements:**

India continues to be dependent on imports to meet the requirement of oilseeds in the country because

1. **Farmers prefer to grow food grains with highly remunerative support prices**
2. **Most of the cultivation of oilseed crops continues to be dependent on rainfall**
3. **Oils from the seeds of tree origin and rice bran have remained unexploited**
4. **it is far cheaper to import oil seeds than to cultivate the oilseed crops**

Which of the statements given above are correct?

(a) 1 and 2 (b) 1, 2 and 3

(c) 3 and 4 (d) 1, 2, 3 and 4

Ans. (b) : India continues to be dependent on imports to meet the requirement of oilseeds in the country due to the fact that agriculture in India is highly dependent on rain. Cost of production of oil seeds is more compared to price offered to them that is why farmers prefer to grow food grains with highly remunerative support prices. India still needs to work upon to extract oil from seeds of tree and rice bran. Every year India pays high amount of foreign exchange to import oil seeds.

70. **Consider the following statements:**

1. **Reserve Bank of India was nationalised on 26 January, 1950.**
2. **The borrowing programme of the Government of India is handled by the Department of Expenditure, Ministry of Finance.**

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : The Reserve Bank of India was nationalised with effect from 1st January, 1949 on the basis of the Reserve Bank of India (Transfer to Public Ownership) Act, 1948. All shares in the capital of the Bank were deemed transferred to the Central Government on payment of a suitable compensation. The finance ministry is likely to set up a full-fledged independent public debt management agency by the end of next year to manage government borrowing programme that runs into lakhs of crore. At present, the government debt, including market borrowings, is managed by the Reserve Bank of India.

PHYSICS

71. **Assertion (A) : In our houses, the Current in A.C. electricity line changes direction 60 times per second.**

Reason (R) : The frequency of alternating voltage supplied is 50 Hz.

(a) Both A and R are individually true and R is the correct explanation of A.

(b) Both A and R are individually true but R is NOT the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

Ans. (d) : An Alternating Current (A.C.) is one which is constantly changing direction. Alternating current have two cycles namely positive half cycle and negative half cycle. Mains electricity is an A.C. supply which is supplied to homes in India at 220 V at 50 Hz. Thus, in our houses the current in A.C. electricity line changes its direction 50 times.

72. **A car is running on a road at a uniform speed of 60 km/hr, the net resultant force on the car is**

(a) driving force in the direction of car's motion

(b) resistance force opposite to the direction of car's motion

(c) an inclined force

(d) equal to zero

Ans. (d) : By Newton's second law F = ma. If a = 0, Fnet = 0 Thus, the resultant force is zero.

73. **A spherical body moves with a uniform angular velocity 'w' around a circular path of radius 'r'. Which one of the following statements is correct?**

(a) The body has no acceleration.

(b) The body has a radial acceleration w^2 r directed towards the centre of the path.

(c) The body has a radial acceleration 2/5 w^2 r directed away from the centre of the path.

(d) The body has an acceleration w^2 r tangential to its path.

Ans. (b) : A body moving with a uniform angular velocity ω on a circular path of radius r has radial acceleration equal to $\omega^2 r$ directed towards the centre of the path.

74. **A weightless rubber balloon is filled with 200cc of water. Its weight in water is equal to**

(a) $\frac{9.8}{5}$N (b) $\frac{9.8}{10}$N

(c) $\frac{9.8}{2}$N (d) Zero

Ans. (d) : In this case, balloon is filled up with water and then put into water. Initially, after some oscillations, it becomes stable because the downward directing force exerted by weight of water in balloon is balanced by the upward directing buoyancy force of water. Hence, the resultant weight in water is zero as there is no resultant force acting upon it.

75. **A person stands at the middle point of a wooden ladder which starts slipping between a vertical wall and the floor of a room, while plane. The path traced by a person standing at the middle point of the slipping ladder is**

(a) a straight line (b) an elliptical path

(c) a circular path (d) a parabolic path

Ans. (c) : The path traced by a person standing at the middle point of the slipping ladder is a circular path.

CHEMISTRY

76. **Consider the following statements:**

1. **Baking soda is used in fire extinguishers.**
2. **Quicklime is used in the manufacture of glass.**
3. **Gypsum is used in the manufacture of Plaster of Paris.**

Which of these statements given above is/are correct?

(a) 1 and 2 (b) 2 and 3

(c) 1 only (d) 1, 2 and 3

Ans. (d) : Baking soda is used as a fire extinguisher for electrical fires and grease fires. When baking soda is heated, it releases carbon dioxide and produces water $2NaHCO_3 \rightarrow H_2O + CO_2 + Na_2CO_3$. Since carbon dioxide is heavier than air and does not support combustion like oxygen does, it smothers the fire while water that is formed cools the fire to below ignition temperature. Calcium oxide, CaO is also called lime, quicklime, or caustic lime. Calcium oxide is widely used in making porcelain and glass. Plaster of Paris is a type of building material based on calcium sulphate hemihydrate nominally $CaSO_4.1/2H_2O$. It is created by heating gypsum to about 300oF (150°C). $2CaSO_4.2H_2O \rightarrow 2CaSO_4.1/2H_2O + 3H_2O$